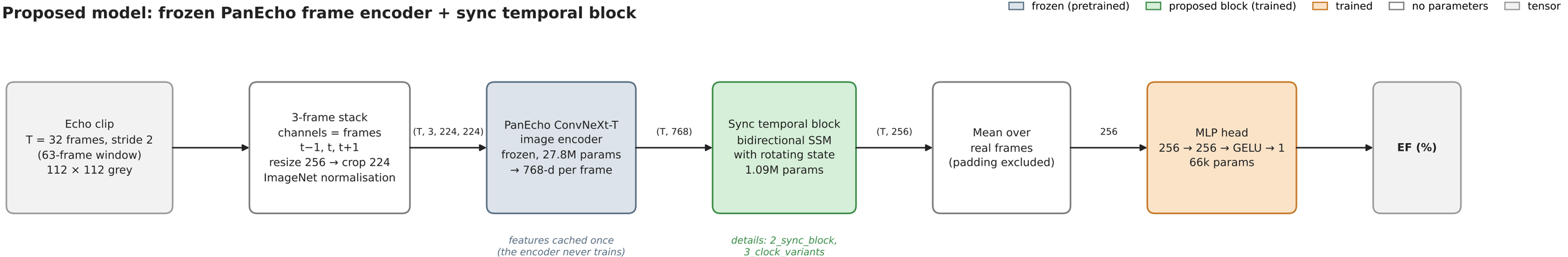
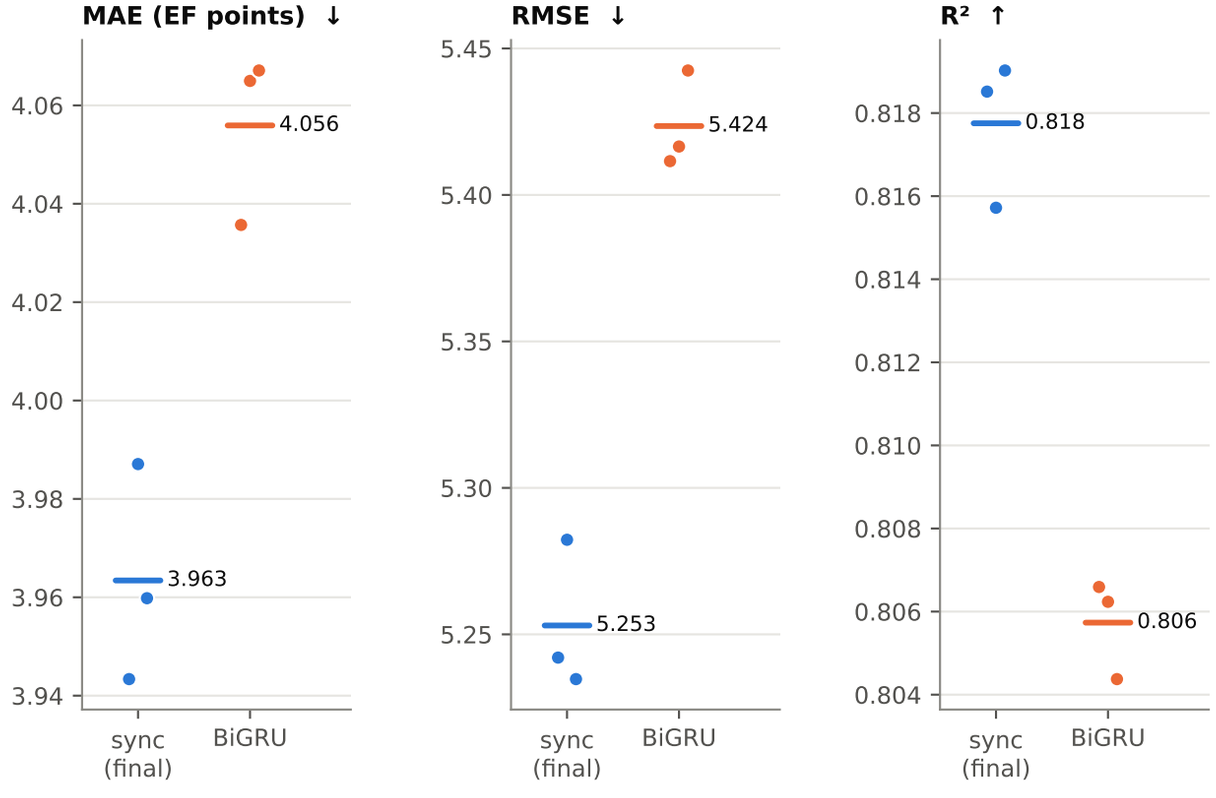
# Proposed model: frozen PanEcho frame encoder + sync temporal block




Training: MSE on standardised EF, AdamW lr 1e-3 (cosine, 5% warm-up), batch 16, 40 epochs; weight EMA 0.999 for validation, checkpoint selection and evaluation.  Evaluation: mean EF over 4 evenly spaced clips per video.

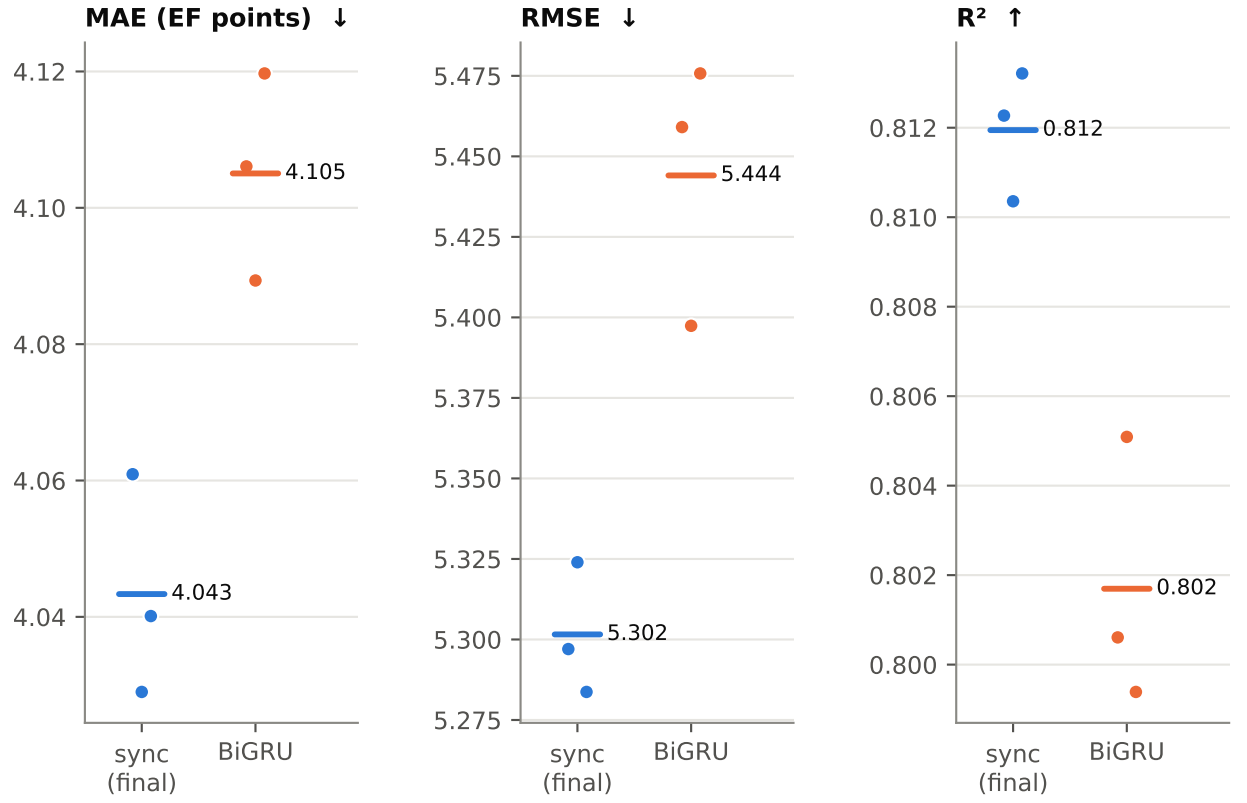
**Final model vs BiGRU — validation set (frozen PanEcho features, weight EMA, 3 seeds, 4 clips per video)**




| metric | sync | BiGRU | better |
|---|---|---|---|
| MAE (EF points) | 3.963 | 4.056 | sync |
| RMSE | 5.253 | 5.424 | sync |
| $R^2$ | 0.818 | 0.806 | sync |
| Pearson r | 0.905 | 0.900 | sync |
| CCC | 0.899 | 0.890 | sync |
| \|bias\| | 0.188 | 0.739 | sync |
| within 5 points (%) | 71.1 | 70.1 | sync |
| within 10 points (%) | 93.7 | 93.0 | sync |
| AUROC, EF < 40% | 0.984 | 0.983 | sync |
| AUROC, EF < 50% | 0.960 | 0.959 | sync |

Means of 3 seeds. Paired over 1288 validation videos (seed-averaged errors):
ΔMAE (sync − BiGRU) = -0.093 [95% CI -0.164, -0.020], p = 0.011

# Final model vs BiGRU — test set (frozen PanEcho features, weight EMA, 3 seeds, 4 clips per video)




| metric | sync | BiGRU | better |
|---|---|---|---|
| MAE (EF points) | 4.043 | 4.105 | sync |
| RMSE | 5.302 | 5.444 | sync |
| $R^2$ | 0.812 | 0.802 | sync |
| Pearson r | 0.902 | 0.900 | sync |
| CCC | 0.894 | 0.886 | sync |
| \|bias\| | 0.444 | 0.995 | sync |
| within 5 points (%) | 70.2 | 69.4 | sync |
| within 10 points (%) | 93.2 | 93.0 | sync |
| AUROC, EF < 40% | 0.983 | 0.981 | sync |
| AUROC, EF < 50% | 0.966 | 0.967 | BiGRU |

Means of 3 seeds. Paired over 1277 test videos (seed-averaged errors):
ΔMAE (sync − BiGRU) = -0.062 [95% CI -0.132, +0.011], p = 0.093